

**University of St. Thomas - Graduate Programs in Software
Foundations of Software Development - Python – SEIS603-02**

Instructor: Syed Ali Naqvi (Call me Ali)

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Date of most recent plan update: 02/01/2026

Class times: 5:30 PM - 8:30 PM (Wednesday)

Room: O'Shaughnessy Science Hall 325 and Online (Please bring your laptops to the class if you possibly can)

Lab times: 8:00 PM to 8:30 PM (during class), however may stretch to 9:00 PM at times

REQUIRED Textbook:

The Quick Python Book, Third Ed.

<https://livebook.manning.com/book/the-quick-python-book-third-edition/about-this-book/>

The Quick Python Book, Second Ed. (2nd Ed is Ok if you cannot get the 3rd edition)

http://128.174.51.21/teaching/SP2019/ASTR414/The_Quick_Python_Book.pdf

RECOMMENDED Books:

How to Think Like a Computer Scientist: Learning with Python 3 Documentation Release 3rd Edition

<https://runestone.academy/runestone/books/published/thinkcspy/index.html>

For further investigation: Online Book - *Dive Into Python 3*

<https://diveintopython3.problemsolving.io/>

HonorLock Requirements: Honorlock is an online proctoring service adopted by University of St. Thomas for use with Distance Learning and testing purposes. As a student you are required to install HonorLock to take your tests and abide by the University policies.

Course Description

This course teaches a comprehensive range of Python language basics using a combination of lectures, in-class labs/demonstrations, and out-of-class homework. The supporting textbooks for this course are freely available online. Language topics included in the course include:

- The way of the program
- Variables, expressions, and statements
- Looping
- Functions, Conditionals, Iteration, Strings, Tuples, Lists, Modules, Recursion
- Object Oriented Thinking using UML
- Basics of Classes and Objects
- Files, Dictionaries, Pickling, Shelving, Simple relational database usage
- Exceptions, Debugging, Unit Testing with PyTest
- LLM's (Large Language Models) for programming will also be introduced for code assistance

After covering Python language basics, the course shifts focus to investigate and discuss current best practice tools and applications of Python in realistic commercial settings. Tools and applications covered include:

Tools:

- Python Integrated Development Environments (IDEs). Multiple IDE's will be covered including **PyCharm and VS Code**.
- The full version of **Anaconda**, a data science tool aggregation environment that includes Python support.
- Also, Jupyter Notebook tool used for Python solution development will be utilized.

Applications:

Students will build many small applications related to each topic discussed in class as well as work on delivering three small projects. All work is done by individuals, and no teamwork is expected in this course.

Course Objectives

This course has the following objectives:

1. Students individually apply best practice software development methods to **learn a comprehensive set of Python language basics**.
2. Students **apply currently recommended, professional quality software development tools** that actively support Python language development methods and best practices taught in this course.
3. Students work individually to code in **Python and develop solutions to given problems**.

At the end of this course, students will be able to:

- Apply basic Python language knowledge and skills acquired in this course to develop Python language programs and applications in the best practice manner.
- Understand and apply basic debugging and unit testing methods in realistic Python development settings.

Pre-Requisites:

- A general understanding of the use and development of computer programs and languages is assumed.

PART 1 - STARTING OUT				Points
Class 1	Wednesday, February 4, 2026	In class and online	Chapter 1. About Python	

			Chapter 2. Getting started	
			Chapter 3. The Quick Python overview	
PART 2 - THE ESSENTIALS				
Class 2	Wednesday, February 11, 2026	Pre-recording Available	Chapter 4. The absolute basics, Understanding Binary Numbers	
Class 3	Wednesday, February 18, 2026	In class and online	Chapter 5. Lists	
			Lab Time + In class Assignment Due or Quiz	3
Class 4	Wednesday, February 25, 2026	In class and online	Chapter 5. Tuples, and sets	
			Lab Time + In class Assignment Due or Quiz	3
Class 5	Wednesday, March 4, 2026	In class and online	Chapter 6. Strings	
			Lab Time + In class Assignment Due or Quiz	3
Class 6	Wednesday, March 11, 2026	In class and online	Chapter 7. Dictionaries	
			Lab Time + In class Assignment Due or Quiz	3
Class 7	Wednesday, March 18, 2026	In class and online	Chapter 8. Control flow	
			Lab Time + In class Assignment Due or Quiz	3
Class 8	Wednesday, March 25, 2026	In class and online	Programming Project 1 Due	10
			Test 1	10
Part 3 - More Advance				
Class 9	Wednesday, April 1, 2026	In class and online	Chapter 9. Functions	
			Lab Time + In class Assignment Due or Quiz	3
Class 10	Wednesday, April 8, 2026	In class and online	Special Topics - Testing Python Code using PyTest	
			Lab Time + In class Assignment Due or Quiz	3
Class 11	Wednesday, April 15, 2026	In class and online	Chapter 12. Using the filesystem	
			Chapter 13. Reading and writing files	
			Lab Time + In class Assignment Due or Quiz	3
Part 4 - OO Programming				
Class 12	Wednesday, April 22, 2026	In class and online	Object-Oriented Programming	
			Lab Time + In class Assignment Due or Quiz	3
			Programming Project 1 Due	10
Class 13	Wednesday, April 29, 2026	In class and online	Chapter 15. Classes and Object-Oriented Programming Chapter	
			14. Exceptions	
			Lab Time + In class Assignment Due or Quiz	3
Class 14	Wednesday, May 6, 2026	In class and online	Chapter 15. Classes and Object-Oriented Programming and Introduction to LLM assistance in code generation	
			Programming Project 3 Due	20
			Test 2	20
			Total	100

Grading for this course is based explicitly on the successful completion and uploading (to the university's Canvas system) of a wide range of specific Python coding assignments and student-completed python code / Jupyter notebooks, as specified in the assignment section for the course as well as completing the tests. Grading will be based on the following grade point scheme: (What you earn is what you get for a grade!)

A = 93%+

A- = 90% - 92%

B+ = 85% - 89%

B = 81% - 84%

B- = 75% - 80%

C = 70% - 74%

F = 0% - 59%

I = (occurs if a student is unable to complete the course - this **must** be negotiated with the instructor)

Department of Software Engineering and Data Science Academic Dishonesty Policy

Honesty and trust among students and between students and faculty are essential for a strong, functioning academic community. Students are expected to complete all academic work on their own. Academic dishonesty is not tolerated in the Department of Software Engineering & Data Science at the University of St. Thomas.

Students are required to complete/submit all academic work as one's own. If a student fails to do so, a grade of zero is issued. If the student commits a second academic dishonesty violation in the same or future course, **a course grade of F is issued**. If a third violation occurs, the student is dismissed from the program.

Common forms of academic dishonesty include, but are not limited to, the following:

- Cheating: Using, accessing, or trying to use unauthorized materials, information, or study aids in any academic work submitted for credit; giving or receiving unauthorized materials or aid in an academic exercise.
- Fabricating or misrepresenting: Falsifying or inventing any information or citation in an academic exercise; fabricating or misrepresenting a reason for an absence or missed deadline; submitting academic work completed largely in one class to satisfy an assignment in a second class without instructor permission; fabricating or tampering with academic records.
- Facilitating Academic Dishonesty: Knowingly helping or attempting to help another student commit academic dishonesty.
- Plagiarizing: Claiming or submitting the ideas, words, or statements of another person as one's own without proper citation regardless of intent to deceive. Even inadvertent plagiarism is a serious matter; students must familiarize themselves with citation standards.

Disability Resources Syllabus Statement: Academic accommodations will be provided for qualified students with documented disabilities including but not limited to mental health diagnoses, learning disabilities, Attention Deficit Disorder, Autism, chronic medical conditions, visual, mobility, and hearing disabilities. Students are invited to contact the Disability Resources office about accommodations early in the semester. Appointments can be made by emailing drscheduling@stthomas.edu. For further information, you can locate the Disability Resources office on the web at: <https://www.stthomas.edu/student-life/resources/disability/>